

One Infrastructure, Infinite Services: Why 5G Slicing Secures the Next Decade

A Demonstration of Kubernetes-Based 5G Standalone (SA) Architecture



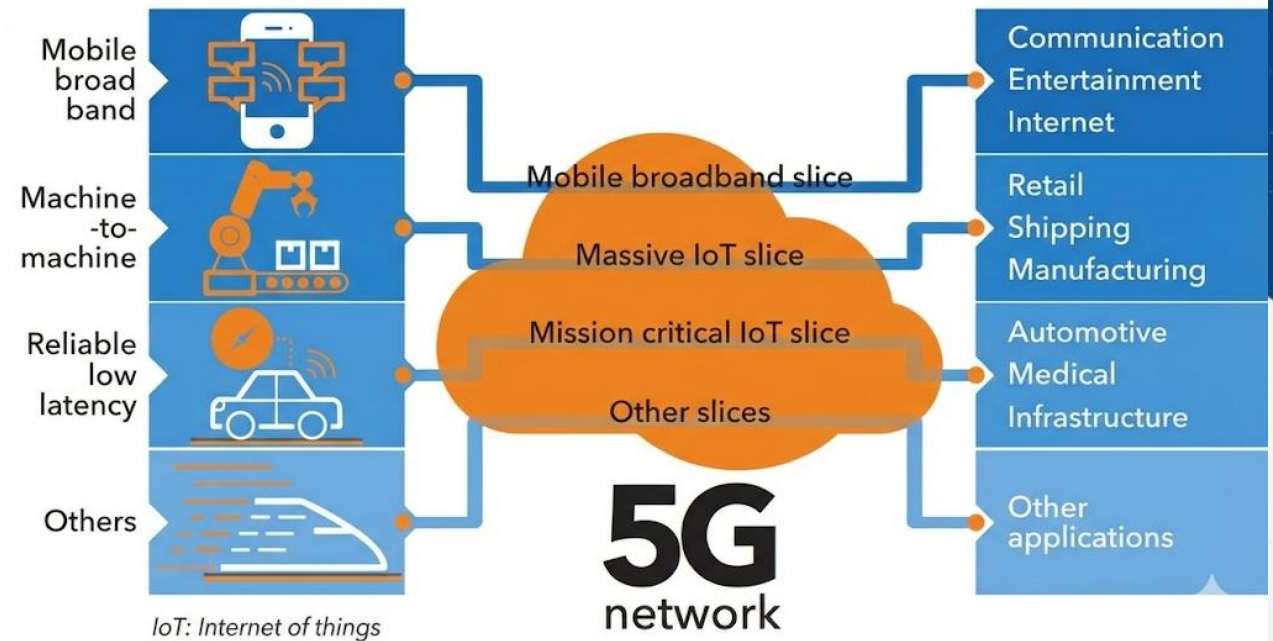
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What is 5G Slicing?

- Definition: Creating multiple virtual networks on top of a single shared physical infrastructure.
- The Shift: Unlike 4G (defined by hardware), 5G is defined by software.
- Legacy (4G): A single lane where a Ferrari (Urgent Data) gets stuck behind a Truck (Netflix).
- 5G Slicing: A multi-lane highway with impenetrable barriers. The heavy truck gets its own lane; the Ferrari speeds through.
- Key Advantage: Hard Isolation. A crash in the public slice cannot affect the industrial slice.

5G network slicing

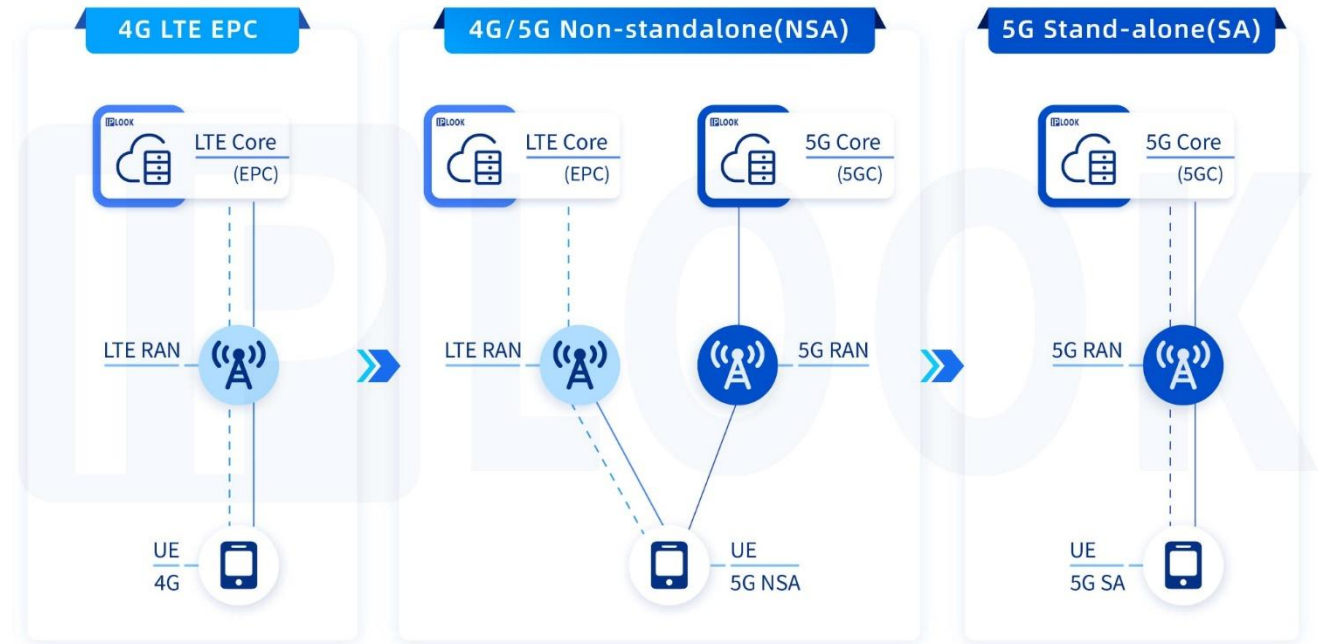
5G network slicing enables service providers to build virtual end-to-end networks tailored to application requirements.



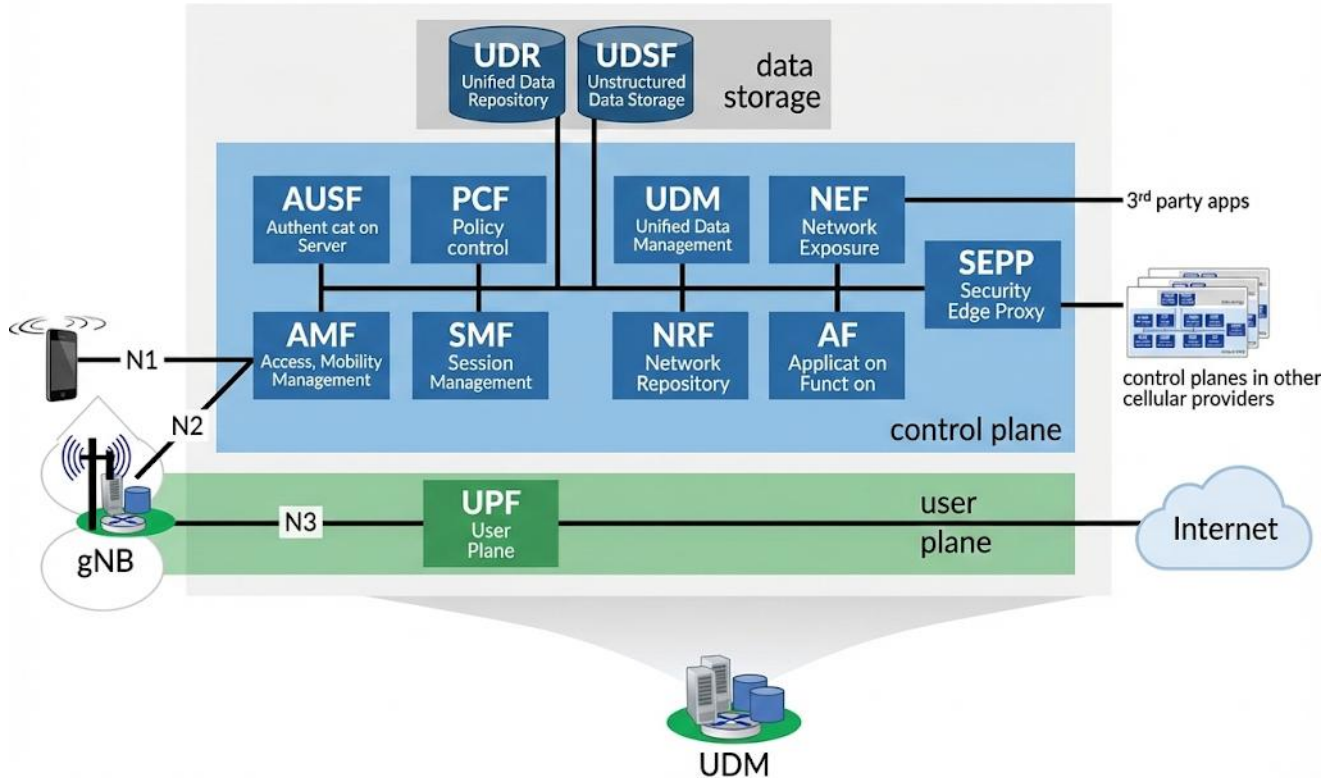
The Evolution (Virtualization & Cloud)

From Hardware Boxes to Cloud Code

- Early 5G (NSA): Fast radio antennas connected to an old 4G Core. No true slicing possible.
- Modern 5G (SA): The Core network is Cloud-Native.
- The Enablers:
 - NFV: Network functions (Firewalls, Routers) became software.
 - Kubernetes: Allows us to deploy the 5G Core as "Pods" (Containers).
- We don't install hardware to add features; we just deploy code.



5G Network Architecture



The list of available updates is more than a week old.
To check for new updates run: `sudo apt update`
New release '24.04.3 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

```
Last login: Tue Jan 20 08:10:15 2026
master@masterServer:~$ kubectl exec -it -n open5gs deploy/srsran-gnb2 -c gnb -- /bin/bash
master@masterServer:~$ kubectl exec -it -n open5gs deploy/srsran-gnb -c gnb -- /bin/bash
root@srsran-gnb-86cf6ffdf-dbg:~# srsran# /srsran/config/start_gnb.sh
```

The PRACH detector will not meet the performance requirements with the configuration {Format 0, ZCZ 0, SCS 1.25kHz, Rx ports 1}.
Warning: Scheduling priority of thread "du_cell#0" not changed. Cause: Not enough privileges.
Lower PHY in executor blocking mode.
Warning: Scheduling priority of thread "phy_worker" not changed. Cause: Not enough privileges.

==== srsran gnb (commit bcf941b34) ===

Connecting to AMF on 10.10.3.200:38412
Available radio types: zmq.
Cell pci=1, bw=20 MHz, dl_arfcn=368500 (n3), dl_freq=1842.5 MHz, dl_ssb_arfcn=368410, ul_freq=1747.5 MHz

==== gNodeB started ===
Type <t> to view trace
t

DL										UL									
pci	rnti	cqi	ri	mcs	brat	ok	nok	(%)	dl_bs	pusch	mcs	brat	ok	nok	(%)	bsr			
1	4601	15	1	0	0	0	0	0%	0	n/a	0	0	0	0	0%	0			
1	4601	15	1	0	0	0	0	0%	0	n/a	0	0	0	0	0%	0			
1	4601	15	1	0	0	0	0	0%	0	n/a	0	0	0	0	0%	0			
1	4601	15	1	0	0	0	0	0%	0	n/a	0	0	0	0	0%	0			
1	4601	15	1	0	0	0	0	0%	0	n/a	0	0	0	0	0%	0			
1	4601	15	1	0	0	0	0	0%	0	n/a	0	0	0	0	0%	0			
1	4601	15	1	0	0	0	0	0%	0	n/a	0	0	0	0	0%	0			
1	4601	15	1	0	0	0	0	0%	0	n/a	0	0	0	0	0%	0			
1	4601	15	1	0	0	0	0	0%	0	n/a	0	0	0	0	0%	0			
1	4601	15	1	0	0	0	0	0%	0	n/a	0	0	0	0	0%	0			
1	4601	15	1	0	0	0	0	0%	0	n/a	0	0	0	0	0%	0			

DL										UL									
pci	rnti	cqi	ri	mcs	brat	ok	nok	(%)	dl_bs	pusch	mcs	brat	ok	nok	(%)	bsr			
1	4601	15	1	0	0	0	0	0%	0	n/a	0	0	0	0	0%	0			

New release '24.04.3 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

```
Last login: Tue Jan 20 08:13:41 2026 from 192.168.56.1
master@masterServer:~$ kubectl exec -it -n open5gs deploy/srsran-ue1 -c ue -- /bin/bash
root@srsran-ue1-68dc56bc7f-9nn6n:~# srue /srsran/config/ue0.conf
Active RF plugins: libsrslan_rf_zmq.so
Inactive RF plugins:
Reading configuration file /srsran/config/ue0.conf...
```

Built in Release mode using commit ec29b0c1f on branch master.

Opening 1 channels in RF device=zmq with args=tx_port=tcp://10.10.3.232:2001,rx_port=tcp://10.10.3.231:2000,base_srate=23.04e6
Supported RF device list: zmq file
CHX base_srate=23.04e6
Current sample rate is 1.92 MHz with a base rate of 23.04 MHz (x12 decimation)
CH0 rx_port=tcp://10.10.3.231:2000
CH0 tx_port=tcp://10.10.3.232:2001
Current sample rate is 23.04 MHz with a base rate of 23.04 MHz (x1 decimation)
Current sample rate is 23.04 MHz with a base rate of 23.04 MHz (x1 decimation)
Waiting PHY to initialize ... done!
Attaching UE...
Random Access Transmission: prach_occasion=0, preamble_index=42, ra-rnti=0x39, tti=334
Random Access Complete. c-rnti=0x4601, ta=0
RRC Connected
PDU Session Establishment successful. IP: 10.41.0.2
RRC NR reconfiguration successful.

Simultaneous Multi-User Access (Hard Slicing)

- Architecture: Dedicated RAN, Shared Core.
- Slice A: Uses dedicated gNB-1 (Worker 1) for isolation.
- Slice B: Uses dedicated gNB-2 (Worker 2) for reliability.
- The "Shared" Element: Both slices are managed by the same Central Core (AMF) on the Master Node.
- Performance:
 - UE 1: Connected to Slice A (IP 10.41.0.2).
 - UE 2: Connected to Slice B (IP 10.41.0.3).
- Significance: We successfully split the radio resources physically while keeping network management centralized.

Additional security updates can be applied when ESM Apps.
Learn more about enabling ESM Apps service at <https://ubuntu.com/esm>

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To check for new updates run: `sudo apt update`
New release '24.04.3 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

```
Last login: Tue Jan 20 08:16:42 2026 from 192.168.56.1
master@masterServer:~$ sudo ip link set enp0s8 promisc on
[sudo] password for master:
master@masterServer:~$ kubectl exec -it -n open5gs deploy/srsran-ue1 -c ue -- /bin/bash
root@srsran-ue1-68dc56bc7f-9nn6n:/# ip netns exec ue0 ip route add default dev tun_srsue
root@srsran-ue1-68dc56bc7f-9nn6n:/# ip netns exec ue0 ping -c 4 10.41.0.1
PING 10.41.0.1 (10.41.0.1) 56(84) bytes of data:
64 bytes from 10.41.0.1: icmp_seq=1 ttl=64 time=769 ms
64 bytes from 10.41.0.1: icmp_seq=2 ttl=64 time=903 ms
64 bytes from 10.41.0.1: icmp_seq=3 ttl=64 time=743 ms
64 bytes from 10.41.0.1: icmp_seq=4 ttl=64 time=832 ms

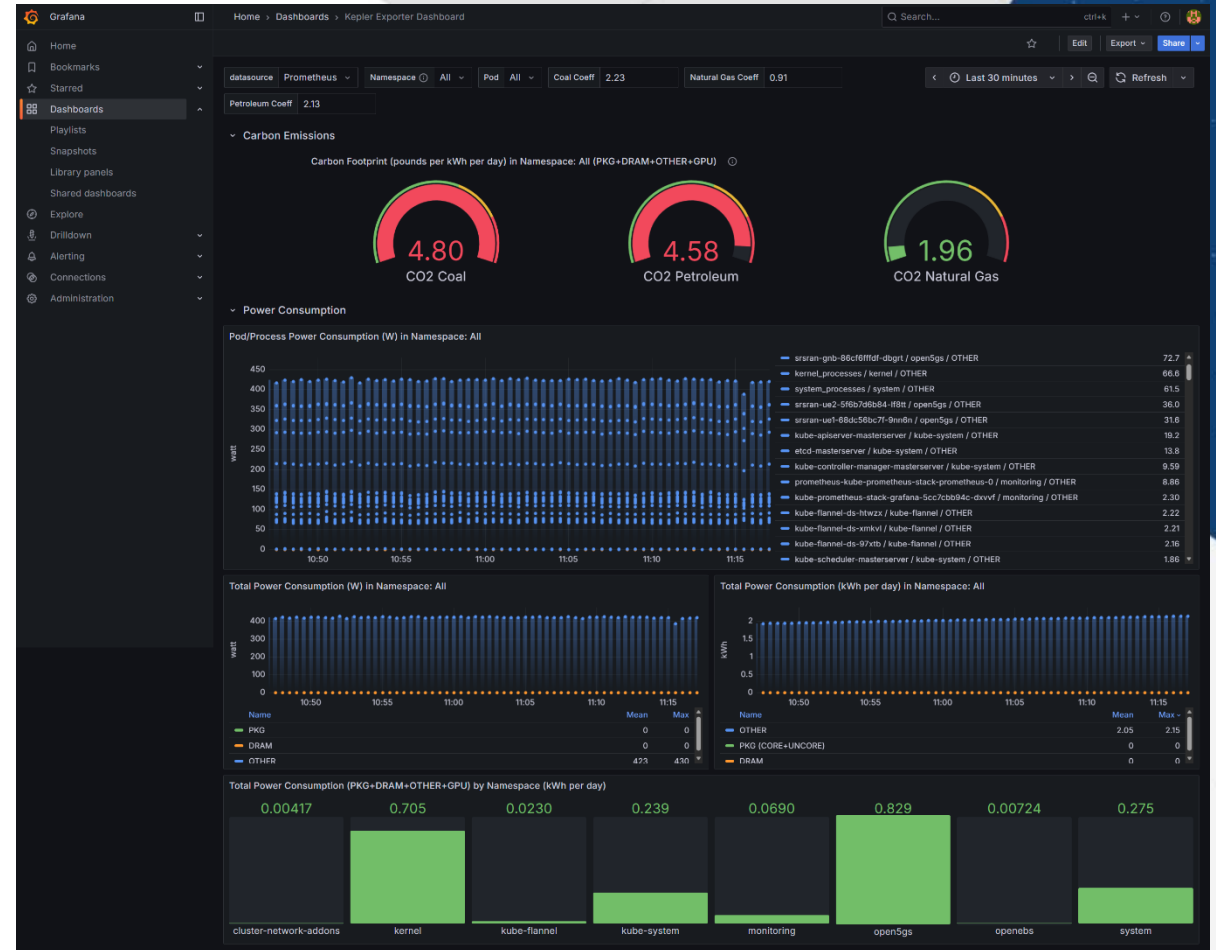
--- 10.41.0.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3014ms
rtt min/avg/max/mdev = 743.311/811.943/903.044/61.672 ms
root@srsran-ue1-68dc56bc7f-9nn6n:/# ip netns list
ue0
root@srsran-ue1-68dc56bc7f-9nn6n:/#
```

```
master@masterServer:~$ sudo ufw disable
sudo iptables -F
[sudo] password for master:
Firewall stopped and disabled on system startup
master@masterServer:~$ kubectl exec -it -n open5gs deploy/srsran-ue2 -c ue -- /bin/bash
root@srsran-ue2-5f6b7d6b84-lf8tt:/# ip netns exec ue0 ip route add default dev tun_srsue
root@srsran-ue2-5f6b7d6b84-lf8tt:/# ip netns exec ue0 ping -c 4 10.41.0.1
PING 10.41.0.1 (10.41.0.1) 56(84) bytes of data:
64 bytes from 10.41.0.1: icmp_seq=1 ttl=64 time=760 ms
64 bytes from 10.41.0.1: icmp_seq=2 ttl=64 time=503 ms
64 bytes from 10.41.0.1: icmp_seq=3 ttl=64 time=711 ms
64 bytes from 10.41.0.1: icmp_seq=4 ttl=64 time=637 ms

--- 10.41.0.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3022ms
rtt min/avg/max/mdev = 503.220/652.855/760.410/96.961 ms
root@srsran-ue2-5f6b7d6b84-lf8tt:/#
```

Dynamic Optimization via Observability

- The Scheduler: Kubernetes monitors node health before assigning a Slice.
- Real-Time Metrics: We use Kepler to track Energy (Joules) and CPU load per Pod.
- Logic:
 - If Node A is power-hungry (Red Zone in dashboard)...
 - The Orchestrator automatically places the new Slice on Node B.
- No single node is overwhelmed; consistent performance is guaranteed.



S-NSSAI & Hard Isolation via Kubernetes

- S-NSSAI: The 32-bit ID tag that routes traffic to the correct slice (e.g., SST: 1 for Video).
- Hard Isolation: We use Kubernetes Namespaces and Resource Quotas.
- Data Proof: The logs show raw throughput (net_rx_bytes) being tracked independently.
- Security: Even if the "Public Slice" floods the network, the "Secure Slice" has reserved CPU/RAM that cannot be touched.

```
master@masterServer:~/ML_Integration_K8s_Scheduler$ cat srsran-gnb2-758db7bf5b-sdcmg_metrics.csv
timestamp,pod_type,cpu_cores,memory_bytes,net_rx_bytes_sec,net_tx_bytes_sec,energy_joules
2026-01-20T09:52:49,RAN,3.2539384052655294,2458271744.0,8948192.546102587,8935061.211839454,0.0
2026-01-20T09:53:05,RAN,3.2885781857230323,2458136576.0,9385239.774251228,9396550.45820295,1191.20699999999367
2026-01-20T09:53:21,RAN,3.303882594863005,2458107904.0,9484739.633149708,9485962.527998064,1190.43300000000773
2026-01-20T09:53:37,RAN,3.31894148849909,2458157056.0,9315067.732267732,9291892.507492507,2358.4649999999674
2026-01-20T09:53:54,RAN,3.30839754458573,2458320896.0,9483862.962550156,9499735.120374499,1189.6980000000033
2026-01-20T09:54:10,RAN,3.308128542892856,2458324992.0,0,0,1164.00599999999357
2026-01-20T09:54:26,RAN,3.2704351611940665,2458066944.0,9095896.721651625,9080925.61469032,1183.0500000000466
2026-01-20T09:54:42,RAN,0.026222402110691014,2458038272.0,0,0,1150.04999999999884
2026-01-20T09:54:58,RAN,0.02339361393323608,2458505216.0,0,0,1177.1039999999992
2026-01-20T09:55:14,RAN,0.024667691809647226,2458374144.0,9211177.97008748,9212670.68008654,1165.57799999999795
2026-01-20T09:55:31,RAN,3.30322193981298,2458382336.0,9120615.84,9143473.68,1200.99900000000107
2026-01-20T09:55:47,RAN,3.2675000757533947,2458324992.0,9676631.640106242,9655440.737051792,1165.0680000000284
2026-01-20T09:56:03,RAN,0.0,2458324992.0,9274748.99430104,9153467.014016291,1188.52799999999912
2026-01-20T09:56:19,RAN,3.247370959433373,2458349568.0,9315887.121488508,9337102.590295512,1152.54300000000051
2026-01-20T09:56:36,RAN,3.3044178220407523,2458185728.0,9244632.886536328,9236902.15999317,1179.58799999999888
2026-01-20T09:56:52,RAN,0.023154065620542126,2458218496.0,9441147.568999402,9442468.512347262,1182.8969999999972
2026-01-20T09:57:08,RAN,3.2362374686957613,2458066944.0,0,0,2345.71200000000577
2026-01-20T09:57:24,RAN,3.368213064772989,2458460160.0,9174105.43189521,9173919.391774297,1211.69399999999595
2026-01-20T09:57:40,RAN,0.02629317820494275,2458501120.0,0,0,1173.05699999999716
2026-01-20T09:57:56,RAN,3.3293038361723566,2458595328.0,0,0,1178.77500000000233
2026-01-20T09:58:13,RAN,3.2492280540005667,2458624000.0,0,0,1161.62400000000107
2026-01-20T09:58:29,RAN,3.28449497830139,2458546176.0,0,0,1177.4909999999998
2026-01-20T09:58:45,RAN,3.236038315952762,2458492928.0,9210413.560170127,9212221.040780585,1158.5549999999993
2026-01-20T09:59:17,RAN,0.02752786885245867,2459013120.0,9410307.792872164,9416183.482066732,1179.49199999999693
2026-01-20T09:59:49,RAN,3.25414141414146,2458664960.0,9398610.069954228,9391326.366698332,1148.8350000000021
2026-01-20T10:00:05,RAN,3.335441915257358,2459066368.0,9179655.099474045,9180667.047793277,1182.65399999999804
2026-01-20T10:00:21,RAN,0.0,2459066368.0,9055653.673973449,9035622.954615623,1190.56500000000023
2026-01-20T10:00:37,RAN,3.2714994166741502,2458988544.0,9367490.793493519,9376294.52563785,2371.61099999999753
2026-01-20T10:00:54,RAN,3.2926502357008904,2459082752.0,9492041.572643638,9486550.078111438,1191.65400000000386
2026-01-20T10:01:10,RAN,0.02448611413240416,2459115520.0,9226066.38276553,9234829.826319305,1157.64599999999497
2026-01-20T10:01:26,RAN,3.293055119132401,2459353088.0,9113870.403808273,9100167.104399212,1186.7400000000049
2026-01-20T10:01:42,RAN,3.277199328143867,2459021312.0,9153260.36193812,9161996.33058127,1162.83899999999781
2026-01-20T10:01:58,RAN,3.1824799280142613,2459398144.0,0,0,1167.35700000000182
2026-01-20T10:02:14,RAN,3.262571389151005,2459295744.0,8952575.161290323,8965567.290322581,1154.83199999999949
2026-01-20T10:02:47,RAN,0.0,2459258880.0,9364533.075513942,9371682.610760566,1171.94399999999595
2026-01-20T10:03:19,RAN,0.025726389709261246,2459181056.0,9053892.570951585,9062566.277128547,1160.41800000000051
2026-01-20T10:03:35,RAN,3.326201943164709,2459430912.0,9131148.758935105,9116851.635053752,1197.00899999999618
2026-01-20T10:03:51,RAN,0.02359204793028439,2459324416.0,9266067.561092477,9282620.268327743,1187.79600000000312
2026-01-20T10:04:07,RAN,0.022091459314052635,2459435008.0,9620428.583075168,9612504.728517855,1178.0760000000001
```



Why This Infrastructure Lasts 15+ Years

- CapEx vs. OpEx: No need to build new towers for 6G; just update the software.
- The Migration:
- Today: 5G Slices for broad coverage.
- Tomorrow: 6G Slices (Terahertz) deployed as microservices on the same cluster.
- Sustainability: Shared hardware means less energy consumption (Green 5G) compared to building separate physical networks.



Conclusion: The Network is Now Code

- 5G Slicing ends the expensive "Rip and Replace" era of telecom. We no longer need to rebuild physical towers to launch new services.
- Because the network is virtualized, it can host future technologies (like a "6G Slice") on the exact same infrastructure we use today.
- We have transformed connectivity from a static "dumb pipe" into a programmable platform. This architecture secures the network's utility for the next decade of innovation.